

# Air Pollution

# Introduction

- Dispersal of pollutants is a very important
- As the atmosphere is not homogeneous
- Pollutants tend to concentrate in specific areas – most of which are near where large human populations reside
- Natural sources are in general more evenly spread, but there are exceptions such as extremely high levels of dust and acidic gases associated with volcanic activity.
- Man's activities (anthropogenic) release heat, gases, aerosols and other wastes into the atmosphere in high conc's overloading the natural dispersal, dilution and recycling systems

# Introduction

- Air pollution definition (WHO)
- *“Air is polluted when one or several pollutants are present in the atmosphere at **such a conc.** and for **so long a time** that they are harmful to man, animals, plants or material property, cause harm or reduce well-being or disturb appreciably its application”.*

# The Atmosphere

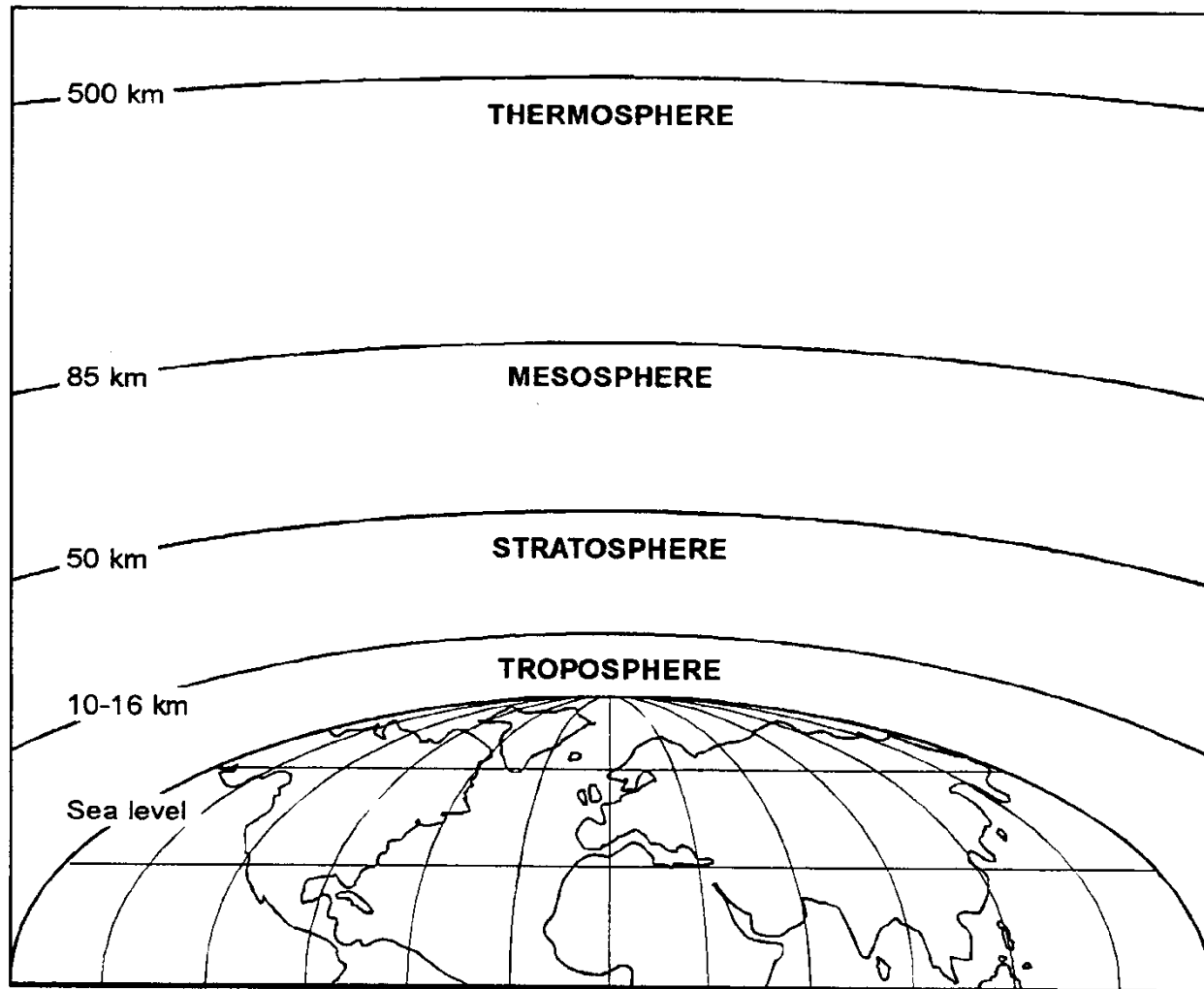
- Earth's atmosphere 160 kilometers deep, 95% of air mass lies within 20 kilometers of the surface.
- Atmosphere is neither uniform, nor static in nature. Its characteristics vary widely with altitude, season, location and solar flare activity.
- Air within a few kilometers of the earth's surface will typically contain the components

| Element          | % (by volume) in the atmosphere | Total Mass in the Atmosphere (x10 <sup>12</sup> tonnes) |
|------------------|---------------------------------|---------------------------------------------------------|
| nitrogen         | 78.08                           | 3900                                                    |
| oxygen           | 20.95                           | 1200                                                    |
| argon            | 0.934                           | 67                                                      |
| carbon dioxide   | 0.035                           | 2.5                                                     |
| Neon             | 0.0018                          | 0.065                                                   |
| Helium           | 0.00052                         | 0.004                                                   |
| Methane          | 0.00015                         | 0.005                                                   |
| Krypton          | 0.0001                          | 0.017                                                   |
| Carbon Monoxide  | 0.00001                         | 0.0006                                                  |
| Ozone            | 0.000002                        | 0.0003                                                  |
| Nitrogen Dioxide | 0.0000001                       | 0.000013                                                |
| Sulfur Dioxide   | 0.0000001                       | 0.000018                                                |
| water            | 0.1 – 5 (normal range 1-3)      | Varies according to location                            |

# The Atmosphere

- The avg. person breathes 20,000L of air per day
- 995 of which is N or O.
- 1% is a mixture of gases and particulates, many of which are pollutants.
- we breathe as much as 200L of pollutants per day!

# The Atmosphere



# Stratification of the Atmosphere

- **stratification** – or layering of the atmosphere
- 95% of the atmosphere's air mass is found in the troposphere. The upper troposphere has a temperature of -56°C.
- At the top of the troposphere is the **tropopause layer** - a barrier to prevent water vapour rising much higher as it causes ice formation.
- Water vapour cannot pass through it.
- **stratosphere**, - The ozone layer is within the stratosphere, and reaches levels of up to 10ppm in the middle of the stratosphere – gets hotter due to this.

# Haze or smog?

- Haze, but how is it different to smog?
- Both typified by a reduction in visibility, but the intensity varies.
- Haze is a condition where the reduction in visibility is not great, and is generally applied to describe the atmospheric conditions over a very large area
- Smog is significant reductions in visibility, generally in metropolitan areas

# Factors that make things BAD

- calm conditions
- low level emission sources
- temperature inversions
- high buildings and narrow streets

# Major Sources of Air Pollution

- Mobile (50 - 70%), and stationary sources.
- 15-25% from heavy industrial stationary sources and as much as 25% from other stationary sources.
- The most important transportation sources at present are major polluters
- Motor vehicles CO, CO<sub>2</sub> H/C's, NO<sub>x</sub> and small amounts of SO<sub>x</sub>
- Motor vehicle exhaust accounts for 40% of all H/C air pollutants and 90% of all NO<sub>2</sub>

# Stack Emissions

- Emission of waste gases, fumes, vapours and smokes to the atmosphere are usually by the use of a smoke stack or chimney.
- stack emission becomes a plume in the atmosphere.
- The plume is an area of concentrated waste emissions that slowly become diluted with the other atmospheric gases.

# Stack Emissions

How dilution happens depends on many factors

- Nature of the waste emission
  - Toxic emissions need to be very dilute
- Volume of the waste
  - Is emission constant or only at certain times in the process.
- Local topography
  - Many cities located in areas surrounded by hills or mountains.
  - low wind and cooler temperatures = photochemical smog.
- Prevailing climate
  - direction of prevailing winds
- The Existing Atmosphere
  - In very polluted cities, more stack emissions not desirable.

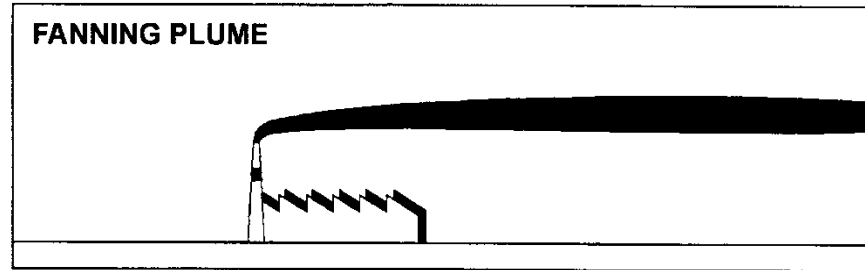
# Plume Behaviour

- Effects of plumes are considered *local* within 500 metres of the stack, and *regional* beyond this.
- Mixing or dispersion of the waste gases and products into the atmosphere = plume behaviour.

# Types of Plumes

- Fanning plumes
- Looping plumes
- Coning plumes
- Fumigating
- Lofting

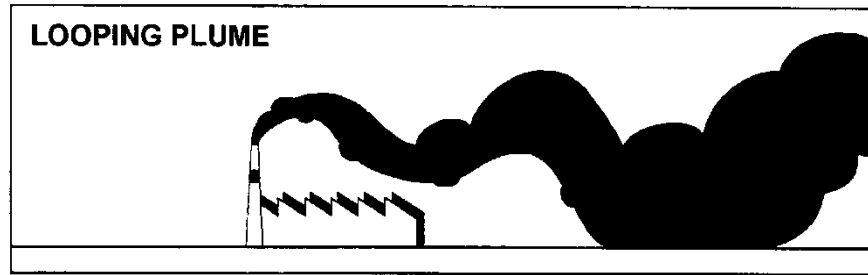
# The Fanning Plume



## ■ Fanning Plumes

- Require stable air and slow vertical movement of the emission
- common after calm clear nights
- temperature inversion limits the rise of the plume into the upper atmosphere
- creates a higher conc. of polluted air at lower levels
- exists for several hours

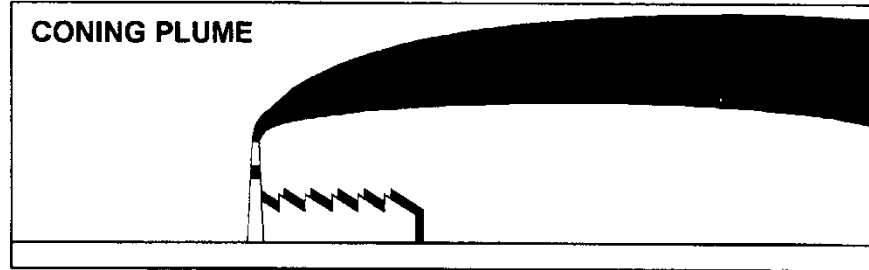
# Looping plumes



## ■ Looping plumes

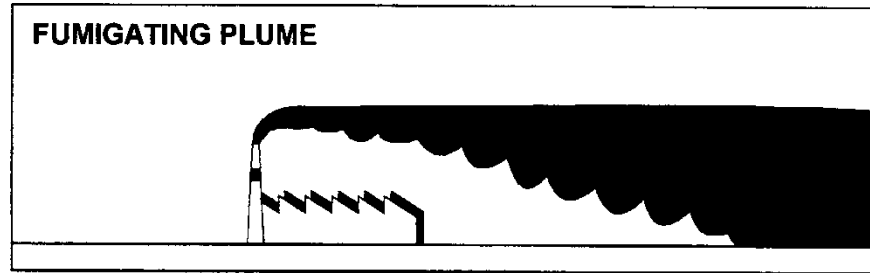
- Require windy conditions which cause the plume can swirl up and down
- common in the afternoon.
- Moderate and strong winds are formed on sunny days creating unstable conditions
- Exists for several hours.

# Coning plumes



- Coning plumes
  - Require moderate winds and overcast days
  - wider than it is deep, and is elliptical in shape
  - exists for several hours.

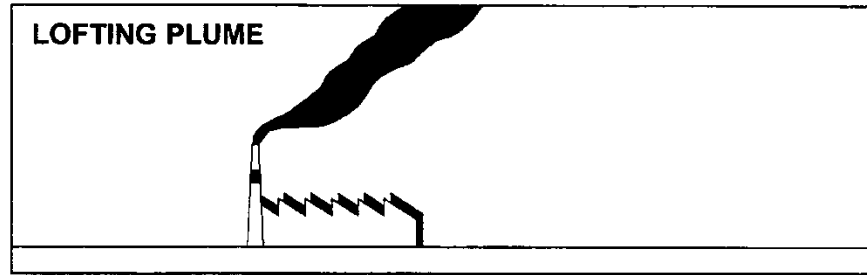
# Fumigating plume



## ■ Fumigating plume

- Is short-lived (fraction of an hour), but reaches the earth's surface.
- occur when the conditions move from stable to unstable
- A fanning plume develops overnight under stable conditions but as the day heats up, unstable air is produced
- unstable air causes the plume to move up and down - can cause localised pollution.

# Lofting plume



- Lofting plume

- When plume is above the inversion layer (or there is no inversion), it becomes a lofting plume.
- Normal wind direction and speed will disperse the plume into the atmosphere without effect from ground warming or cooling.

# Stack emissions

- factors used to establish the amount of stack emission allowed, and its conc. to the atmosphere include:
  - smoke stack (chimney) height,
  - local topography,
  - temperature,
  - emission rates,
  - chemical reactivity, and
  - existing air pollution problems
  - wind allow rapid dispersal of pollutants.

# Primary vs Secondary pollutants

- Not all of the pollutants found in the atmosphere are the direct result of emissions.
- Many pollutants arise from chemical reactions in the atmosphere with other substances or light (photochemical reactions).
- Pollutant substances that are directly emitted into the atmosphere = *primary pollutants*.
- Substances not directly emitted into the atmosphere, formed by chemical reactions in the atmosphere = *secondary pollutants*.

# Particulate Pollutants

- Very small *solid* or *liquid* particles
- Individual particles may vary in size, geometry, chemical composition and physical properties
- May be of natural origin (pollen or sea spray) or man made (dust, fume and soot)
- Provide a reactive surface for gases and vapours in the formation of secondary pollutants
- Particles also diffuse light reducing visibility
- Come from stack emissions, dusty processes, unsealed roads, construction work and many other sources

# Particulate Pollutants

- **Dusts**

- large solid particles

- **Fume**

- solid particles (metallic oxides) formed by condensation of vapours from a chemical reaction process or physical separation process

- **Mist**

- liquid particles formed by condensation of vapours or chemical reaction.



- **Smoke**

- solid particles formed as a result of incomplete combustion of carbonaceous materials.

- **Spray**

- a liquid particle formed by the atomisation of a parent liquid.

# Particle Size

- Particles range in size from 0.005 - 500 $\mu\text{m}$ .
- Smallest of these are clusters of molecules whilst the largest are easily visible with the naked eye.
- Sizes given are not the physical size, but rather the aerodynamic equivalent diameter which relates the particle to the behaviour of an equivalent spherical particle.

# Particle Size

- Particles less than  $1\mu\text{m}$  in diameter behave like gases (remain suspended, may coalesce, move in fluid streams),
- Larger particles act like solids (affected by gravity, don't stay suspended long, don't coalesce).
- Smaller particles generally derive from chemical reactions, whereas the larger particles ( $10\mu\text{m}$  or greater) are usually generated mechanically and tend to be basic.
- Smaller particles most dangerous to health
- In urban areas there is an approx. even distribution between fine and coarse particles, this is weather dependent.
- Fine particulate matter spread over much greater distances

# Particle behaviour in the atmosphere

- Particles can undergo many physical and chemical changes;
  - grow in size,
  - absorb or desorb gases from their surfaces,
  - change electrical charge,
  - collide or adhere with other particles,
  - absorb water.
- changes the particle size and affect its atmospheric lifetime.

# Total Suspended Particles (TSP)

- Most particles concentrated into three main size groups
- Larger particles around  $10\mu\text{m}$  in size
- Smaller particles in size groups centred around  $0.2$  and  $0.02\mu\text{m}$ .
- Only particles of  $<10\mu\text{m}$  penetrate into the human lung
- Analyse air for only this fraction to estimate its potential danger to human health =  $\text{PM}_{10}$  sampling.
- Particles  $<2.5\mu\text{m}$  in size can penetrate deep into the lung tissue and are especially dangerous =  $\text{PM}_{2.5}$  sampling